

MA587 HW 3

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Question 1 Second order Problem

1(a) First, the differential equation is multiplied by test vector $v \in V$ and integrated over the domain,

$$-\int_0^1 u'' v dx = \int_0^1 f v dx \quad (1)$$

Gauss Divergence theorem is used to reduce the left-hand side,

$$-\left[\cancel{u'v} \Big|_0^1 - \int_0^1 u'v' dx \right] = \int_0^1 f v dx \quad (2)$$

Note: In the original image, there is a handwritten arrow pointing from the text $v(0)=v(1)=0$ to the boundary terms $\cancel{u'v}|_0^1$ in the equation above.

which yields the final weak form,

$$a(u, v) = L(v) \quad \forall v \in V \quad (3)$$

$$a(u, v) = \int_0^1 u'v' dx \quad (4)$$

$$L(v) = \int_0^1 f v dx \quad (5)$$

1(b) To show that a is symmetric, note that the multiplication inside the integral is commutative, so

$$a(u, v) = \int_0^1 u'v' dx = \int_0^1 v'u' dx = a(v, u) \quad (6)$$

and thus a is symmetric. To show that a is continuous, first note that

$$\|u\|_{H_0^1}^2 = \|u\|_{L^2}^2 + \|u'\|_{L^2}^2 \quad (7)$$

$$\|u'\|_{L^2}^2 \leq \|u\|_{H_0^1}^2 \quad (8)$$

$$\|u'\|_{L^2} \leq \|u\|_{H_0^1} \quad (9)$$

and also

$$\|u\|_{L^2} \leq \|u\|_{H_0^1} . \quad (10)$$

By the Cauchy-Schwarz (CS) inequality,

$$\int_0^1 u'v' dx = \langle u', v' \rangle_{L^2} \leq \|u'\|_{L^2} \|v'\|_{L^2} , \quad (11)$$

and thus, using Equation 9,

$$\int_0^1 u'v' dx \leq \|u\|_{H_0^1} \|v\|_{H_0^1} , \quad (12)$$

and a is continuous. To show $L(v)$ is continuous, apply the CS inequality,

$$\int_0^1 f v dx = \langle f, v \rangle_{L^2} \leq \|f\|_{L^2} \|v\|_{L^2} \quad (13)$$

and apply Equation 10,

$$\int_0^1 f v dx \leq \|f\|_{L^2} \|v\|_{H_0^1} . \quad (14)$$

1(c) To show that a is coercive, consider that

$$|v(x)| = \left| \cancel{v(0)} + \int_0^x v'(s) ds \right| \quad (15)$$

$$|v(x)| \leq \int_0^1 |v'(s)| ds = \langle 1, v' \rangle_{L^2} . \quad (16)$$

Then, using the CS inequality,

$$|v(x)| \leq \|1\|_{L^2} \|v'\|_{L^2} \quad (17)$$

$$\|v(x)\|_{L^2} \leq \|(\|v'\|_{L^2})\|_{L^2} \quad (18)$$

$$\|v(x)\|_{L^2} \leq \|v'\|_{L^2} \quad (19)$$

1(d)

Question 2 Poincare Inequality

2(a) Similarly to the previous problem, and using the notation $x_1 = x, x_2 =$

$$v(x, y) = \cancel{v(0, y)} + \int_0^x \frac{\partial v(s, y)}{\partial x} ds \quad (20)$$

$$v(x, y)^2 = \left(\int_0^x \frac{\partial v(s, y)}{\partial x} ds \right)^2. \quad (21)$$

Then, by the CS inequality,

$$v(x, y)^2 \leq \int_0^x \left(\frac{\partial v(s, y)}{\partial x} \right)^2 ds. \quad (22)$$

Since the term inside the integral is strictly non-negative,

$$v(x, y)^2 \leq \int_0^1 \left(\frac{\partial v(s, y)}{\partial x} \right)^2 ds. \quad (23)$$

Then, since $\|\nabla v\|^2 = \left(\frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2$ and thus $\frac{\partial v}{\partial x}^2 \leq \|\nabla v\|^2$,

$$v(x, y)^2 \leq \int_0^1 \|\nabla v\|^2 dx. \quad (24)$$

Integrating this once more over the domain,

$$\int_0^1 v(x, y)^2 \leq \int_0^1 \int_0^1 \|\nabla v\|^2 dx dx \quad (25)$$

$$\int_0^1 v(x, y)^2 \leq \int_0^1 \|\nabla v\|^2 dx. \quad (26)$$

2(b) Note that $\|v\|_{H^2}^2 = \|v\|_{L^2}^2 + \|\nabla v\|_{L^2}^2 + \|\Delta v\|_{L^2}^2$. Begining with the $\|v\|_{L^2}^2$ term, as in the previous part,

$$v(x, y) = \cancel{v_x(\theta, y)} + \int_0^x v_x(s, y) ds \quad (27)$$

$$v(x, y)^2 = \left(\int_0^x v_x(s, y) ds \right)^2. \quad (28)$$

Similarly,

$$v_x(s, y) = \cancel{v_x(x, \theta)} + \int_0^y v_{xy}(s, q) dq, \quad (29)$$

so, plugging this into Equation 28,

$$v(x, y)^2 = \left(\int_0^x \int_0^y v_{xy}(s, q) ds dq \right)^2 \quad (30)$$

Applying the CS inequality twice,

$$v(x, y)^2 \leq \int_0^x \int_0^y v_{xy}(s, q)^2 ds dq. \quad (31)$$

Since the term inside the integral is strictly non-negative,

$$v(x, y)^2 \leq \int_0^\Gamma \int_0^\Gamma v_{xy}(x, y)^2 dx dy. \quad (32)$$

where Γ is the length of the square. Applying Green's formula and using the boundary conditions,

$$v(x, y)^2 \leq \int_\Omega v_{xx} v_{yy} d\mu \leq \int_\Omega |v_{xx}| |v_{yy}| d\mu. \quad (33)$$

Since

$$|\Delta v|^2 = |v_{xx}|^2 + 2|v_{xx}| |v_{yy}| + |v_{yy}|^2, \quad (34)$$

and thus $|v_{xx}| |v_{yy}| \leq \frac{1}{2} |\Delta v|^2$,

$$v(x, y)^2 \leq \frac{1}{2} \int_\Omega |\Delta v|^2 d\mu. \quad (35)$$

Integrating over the domain yields

$$\int_{\Omega} v(x, y)^2 d\mu \leq \frac{1}{2} \Gamma^2 \int_{\Omega} |\Delta v|^2 d\mu, \quad (36)$$

where again Γ is the length of the square. Next, to show the inequality for $\|\nabla v\|_{L^2}^2$, again apply the expansion in Equation 34,

$$|\nabla v|^2 = |v_{xx}|^2 + |v_{yy}|^2 \leq |\Delta v|^2, \quad (37)$$

then integrate over the domain to yield

$$\int_{\Omega} |\nabla v|^2 d\mu \leq \int_{\Omega} |\Delta v|^2 d\mu. \quad (38)$$

Then, combining the inequalities Equation 36 and Equation 38,

$$\|v\|_{H^2} = \|v\|_{L^2} + \|\nabla v\|_{L^2} + \|\Delta v\|_{L^2} \leq (\Gamma^2 + 2) \int_{\Omega} |\Delta v|^2 d\mu. \quad (39)$$

Question 3 Coercivity

3(a) Beginning with the differential form, multiply by $v \in H^1$ and integrate over the domain,

$$-\int_{\Omega} \nabla^2 u \cdot v d\mu = \int_{\Omega} f v d\mu \quad (40)$$

then by the Gauss Divergence Theorem,

$$-\left[-\int_{\Omega} \nabla u \nabla v d\mu + \int_{\Gamma} (\nabla u \cdot \mathbf{n}) v dS \right] = \int_{\Omega} f v d\mu. \quad (41)$$

Using the boundary condition to evaluate the second term,

$$\int_{\Omega} \nabla u \nabla v d\mu - \int_{\Gamma} (g - \gamma u) v dS = \int_{\Omega} f v d\mu. \quad (42)$$

By rearranging terms, the weak form is obtained,

$$a(u, v) = L(v) \quad \forall v \in H^1 \quad (43)$$

$$a(u, v) = \int_{\Omega} \nabla u \nabla v d\mu + \gamma \int_{\Gamma} uv dS \quad (44)$$

$$L(v) = \int_{\Omega} f v d\mu + \int_{\Gamma} g v dS. \quad (45)$$

3(b) To show that a is symmetric, consider multiplicative commutativity (as with problem 1),

$$a(u, v) = \int_{\Omega} \nabla u \nabla v d\mu + \gamma \int_{\Gamma} uv dS = \int_{\Omega} \nabla v \nabla u d\mu + \gamma \int_{\Gamma} vu dS = a(v, u). \quad (46)$$

To show that a is continuous, first note that

$$\|u\|_{H^1}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2. \quad (47)$$

3(c)

Question 4 Biharmonic Problem

4(a)

4(b)